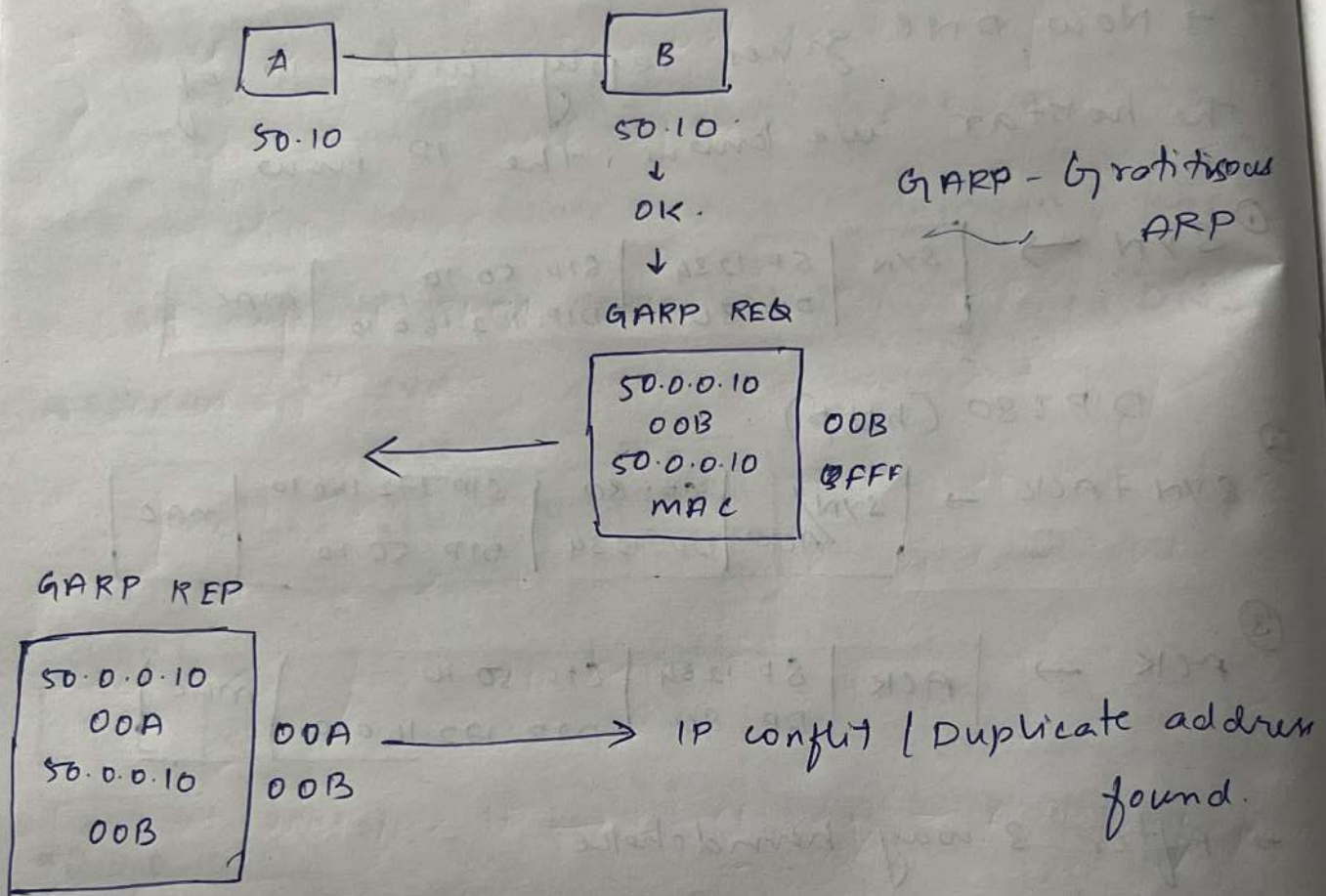


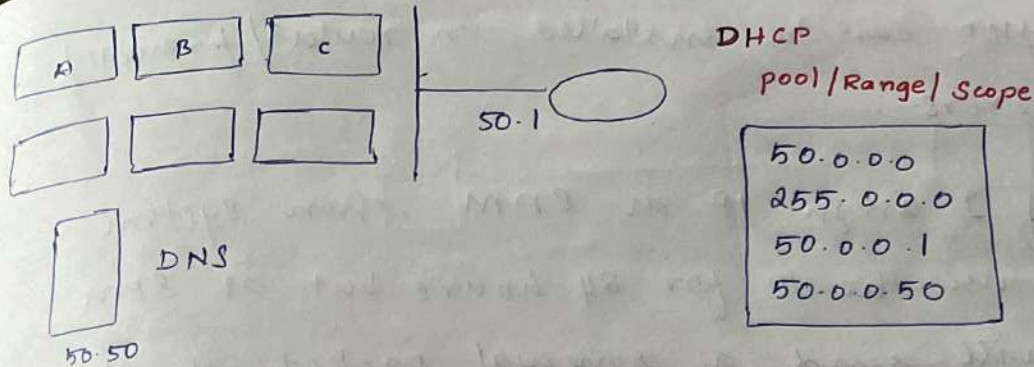
Day-17:

→ If we assign same IP to directly connected devices, then an error comes

IP conflict / Duplicate address found.



* Once we do IP configuration for a device, it will send GARP Req. If it gets a reply then error. If it doesn't get a reply then no problem.



→ We cannot give IP config. for all devices (Eg: 600) we can use DHCP.

DHCP - Dynamic host configuration Protocol

DHCP → First come First Serve.

If a client asks for IP, then only a DHCP will assign IP. Once it assigns IP, DHCP will note the IP & Device ^{mac} ~~name~~ in DHCP table.

When a client sends DHCP request, it will send ARP request first. Only when there is no reply it will assign that particular IP or else it will send next IP.

DHCP TABLE

50.0.0.1 - A
50.0.0.2 - B

lease → 24 hours

Renewal time 12 hours.

After 24 hrs. automatically IP will change if IP is not in use.

At renewal time → it will check whether the PC is ON by sending a packet

* DHCP can be installed in routers/firewall/servers.

* If I assign IP at 8AM, then system can use the IP for 24 hours but at 8PM I will send a renewal packet if it replies then I will update the lease hours again 24 hours.

* But if at 8PM there is no reply then at 8AM IP will be removed automatically.

* If at 7AM PC is ON then at that time packet will be sent, it will update the lease time for 24 hours.

If client asks for IP → DHCP req. → ARP req. from router
→ if gets reply: It will not assign that IP.
→ if no reply: then that IP is not assigned already, so it will give that IP.

* For important devices, only static IP should be given because it should be permanent.

A B C D
 50.1 50.2 50.3 50.4

↳ Static route.

DHCP will send ARP request everytime it assigns IP, so if it asks 50.3 it will get reply and it will not assign.

→ But when sending request the PC is off then no reply, it will assign that IP to D.

→ To avoid such errors while we are giving DHCP pool we have to give excluded also.

DHCP POOL

50.0.0.0
 255.0.0.0
 50.0.0.1
 50.0.0.50

Excluded

50.1-50.100

→ These IP can be used for giving static IP for imp. devices.

DHCP: - * Service used to provide IP address dynamically to client.

* Works with UDP protocol

* DHCP client port no: 68

DHCP service port no: 67